

Labor market

The marginal revenue productivity theory (MRP):

Marginal revenue productivity theory suggests that wage differentials result from difference in labor productivity and the value of the output that the labor input produces. The MRP theory outlined below is based on the assumption of a perfectly competitive labor market and rests on a number of key assumptions that realistically are unlikely to exist in the real world. Most of our labor markets are imperfect - this is one of the many reasons for the existence and persistence of large earnings differentials between occupations.

- Workers are homogeneous in terms of their ability and productivity
- Firms have no buying power when demanding workers (i.e. they have no monopsony power)
- Trade unions have no impact on the available labor supply
- The physical productivity of each worker can be accurately and objectively measured and the market of the output produced by the labor force can be calculated.

The industry supply of labor is assumed to be perfectly elastic. Workers are occupationally and geographically mobile and can be hired at a constant wage rate.

The marginal revenue which the firm gets from employing one more worker is called MRP. It is the value of the marginal product of labor. The derived nature of the demand curve implies that the strength of the demand for labor will depend on:

1. The productivity of labor in helping to produce a good
2. The market value or price of the good which the labor is producing.

MRP is calculated by multiplying the marginal physical product (MPP) of labor and the marginal revenue (MR) gained by selling one more unit of output.

$$\text{MRP of labor} = \text{MPP of labor} \times \text{MR}$$

If the firm is operating under perfect competition, the price is constant, so $P = \text{MR}$ and the MRP under perfect competition will be

$$\text{MRP} = \text{MPP} \times P$$

There is no such thing as luck, believe in hard work

Labor market

The MP curve follows the law of diminishing returns i.e. the MP increases due to increasing returns and falls due to decreasing returns. The MRP curve is the product of MP and a constant MR, so the shape of MRP resembles the shape of MP curve.

The ARP curve is the product of average product and the constant MR. the ARP shows the value of output per unit of labor. The relationship between ARP and MRP is the same as the relationship between MP and AP i.e. the MRP intersects the ARP at ARP's maximum point.

The marginal cost of labor: It is the cost of employing an extra unit of labor. Under perfect competition the wage rate is constant. So MC of labor = W .

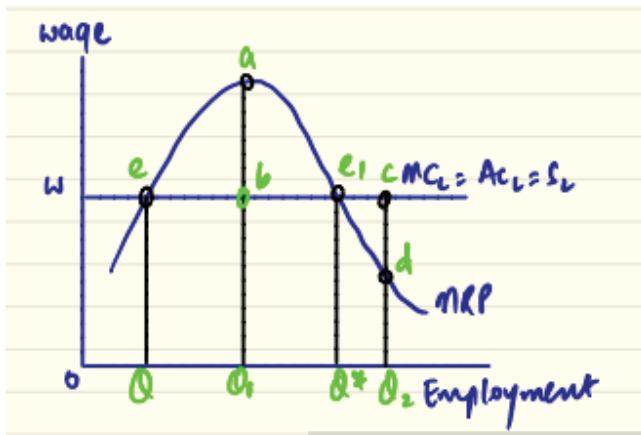
The profit maximizing firm will employ additional workers as long as those workers are adding more to the firm's revenue than to the firms' costs. In other words, labor will be employed up to the point where its MRP is equal to the marginal cost or wage rate. $MRP = W$.

If the number of workers a firm is currently hiring is such that the MRP of the last worker exceeds his wages, the firm can clearly profit by hiring more workers. By doing so the MPP of the workers will fall because of the diminishing return. This will reduce the MRP and the producer will continue to hire the labors until $MRP = W$. similarly, if the number of labors hired is such that the wages exceed the MPP, the producer will lay-off some of the workers causing the MRP of the workers to rise until $MRP=W$.

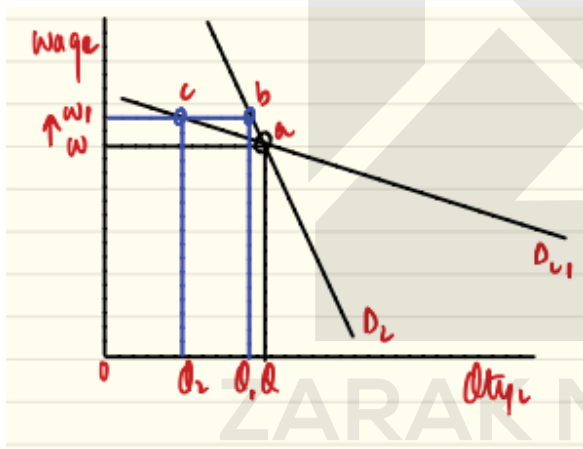
The profit maximizing level of employment: The profit maximizing firm will employ additional workers as long as those workers are adding more to the firm's revenue than to the firm's costs. In order words, labor will be employed up to the point where its MRP is equal to the marginal cost or the wage rate as shown in the figure.

There is no such thing as luck, believe in hard work

Labor market



Elasticity of demand for labour: Elasticity is an important concept in labour market economics. In this case we are concerned with how responsive is the demand for labour as a factor input in production when there is a change in the ruling market wage rate. Will businesses cut back aggressively on employment if wage rates increase? Will the number of people taken on expand if labour becomes cheaper relative to other factor inputs?



1. The proportion of labour costs in the total costs of a business: When businesses' labour expenses are a high proportion of total costs, labour demand can be expected to be more elastic than a business or industry where fixed costs of capital are the dominant business expense

2. The ease and cost of factor substitution: Can a firm substitute between labour and capital when the relative prices of each change over time? Can the main car producers reduce their labour demand and replace it with more capital? The answer is yes, although there are naturally costs involved (including the financial

There is no such thing as luck, believe in hard work

Labor market

costs of redundancies and time lags in bringing in new capital inputs). If wage rates in the hotel and leisure sector increase following the introduction of the national minimum wage, can hoteliers or catering businesses afford to reduce their demand for labour?

The demand for labour tends to be more elastic when labour and capital are easily substitutable. This depends on the nature of the production process, the added "human" value that the labour input provides (particularly in service industries) and the flexibility of the labour market (for example the ease and cost of hiring & firing labour is influenced by existing employment laws). When labour is considered a necessity in the production process, the demand will be inelastic in response to wage changes

3. The price elasticity of demand for the final output produced by a business:

If a firm is operating in a highly competitive market where final demand for the product is price elastic, they may have little power to pass on higher wage costs to consumers through a higher price, the demand for labour may therefore be more elastic as a consequence

4. The time period under consideration: In the short run, at least one factor of production is assumed to be fixed so the demand for labour as an input will be more inelastic compared to the long run when a business has a much greater opportunity to vary the factor mix between labour and capital

The supply of labor: It is a measure of the number of hours of work which is offered at a given wage rates over a given period of time. It is determined by the number of workers and the average number of hours each worker is prepared to offer. The number of people willing to work is called the labor force. The total supply of labor depends on the following factors:

- **The size and age composition of population:** The total population in an economy sets an upper limit to the supply of labor. The greater is the size of total population, the more are the chances of a large supply of labor. The age composition of a population takes account of the proportions in different age groups. Two countries might have the same total populations but very different age compositions and hence very different numbers in the working age groups e.g. in developing countries much of the population lies within the age group 0 to 15 years. So the size of the labor force will be small as

There is no such thing as luck, believe in hard work

Labor market

compared to the developed country of the same population because most of the population lies between 15 to 60 years of age.

- **Participation rate:** The proportion of the total population that is willing to work is called labor force participation rate. This varies in response to many influences, for example changes in attitudes and tastes. The enormous rise in female participation rates over the last few decades has significantly increased the labor force in many countries. Generally, a rise in the demand for labor and an accompanying rise in earnings will lead to an increase in the proportion of the population willing to work.
- **Hours worked:** Since the number of hours offered for work at a particular wage rate is the measure of the labor supply, the greater the number of hours offered the greater is the supply of labor. The number of hours offered for work primarily depends on the wage rate, the greater is the wage rate the more will be the supply of labor as the labor gives up some of his leisure for income and more people will tend to take employment.

Supply of labor in a particular occupation/industry: The supply of labor to a particular occupation is influenced by a range of monetary and non-monetary considerations.

1. The real wage rate on offer in the industry itself - higher wages raise the prospect of increased factor rewards and should boost the number of people willing and able to work
2. Overtime: Opportunities to boost earnings come through overtime payments, productivity-related pay schemes, and share option schemes and financial discounts for employees in a certain job.
3. Substitute occupations: The real wage rate on offer in competing jobs is another factor because this affects the wage and earnings differential that exists between two or more occupations. So for e.g. an increase in the relative earnings available to train plumbers and electricians may cause some people to switch their jobs.
4. Barriers to entry: Artificial limits to an industry's labor supply (e.g. through the introduction of minimum entry requirements or other legal barriers to entry) can restrict labor supply and force average pay and salary levels higher - this is particularly the case in professions such as legal services and medicine where there are strict 'entry criteria' to the professions. Indeed these labor market barriers are partly designed to keep pay levels high as well as being methods of maintain the quality of people entering these

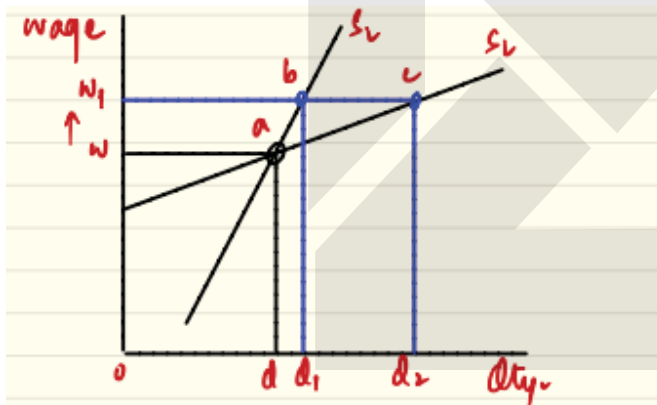
There is no such thing as luck, believe in hard work

Labor market

professions. The length of time it takes to qualify as an architect restricts the labor supply and keep salaries high.

5. Improvements in the occupational mobility of labor: For e.g. if more people are trained with the necessary skills required to work in a particular occupation.
6. Non-monetary characteristic of specific jobs - these can be important - they include factors such as the level of risk associated with different jobs, the requirement to work anti-social hours or the non-pecuniary benefits that certain jobs provide including job security, opportunities for promotion and the chance to live and work overseas, employer-provided in-work training, employer-provided or subsidized health and leisure facilities and other in-work benefits including occupational pension schemes.

Elasticity of supply of labor: It measures the responsiveness of the labor supplied to the changes in the wage rate. The supply of labor is elastic when a given percentage change in the wage rate causes a greater % change in the supply of labor and it will be inelastic when the % change in labor supply is relatively smaller.



The main **determinants** of the elasticity of supply of labor in an industry are:

- **The level of employment:** If there is a high unemployment the supply of labor will be relatively elastic. A small rise in the wage rate is likely to attract a large number of job applicants.
- **The mobility of labor:** The more occupationally and geographically mobile labor is, the more elastic the supply of labor will be.
- **The length of training:** The longer it takes to train a worker, the more inelastic supply will tend to be. For instance, it takes seven years to train a

There is no such thing as luck, believe in hard work

Labor market

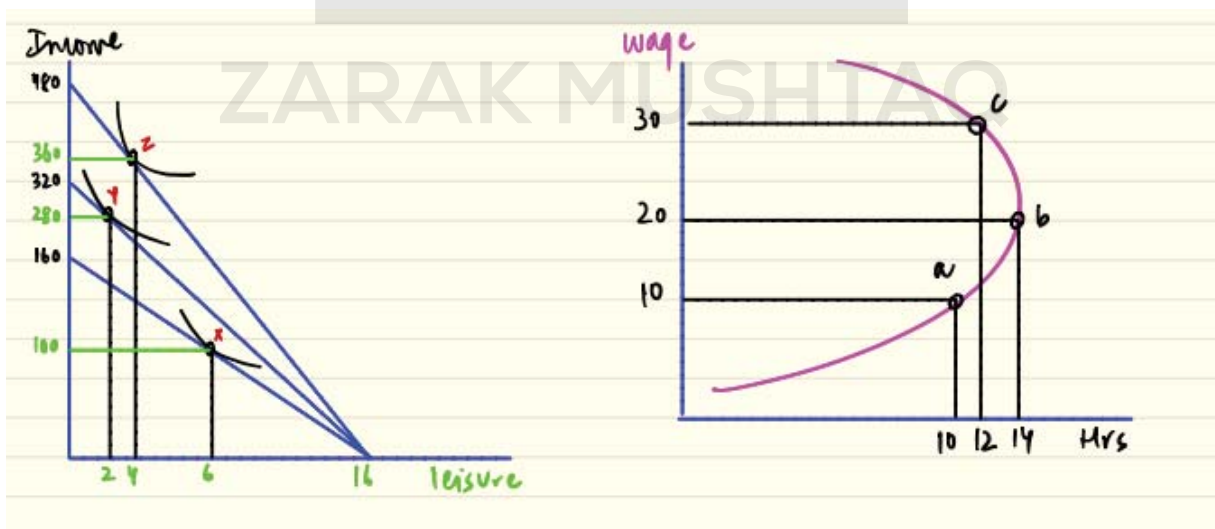
surgeon, so a rise in wages of surgeons will take some time before a rise in the number of surgeons.

- The qualifications and skills required: The higher the skills and qualifications required the lesser the number of workers which qualify for the job making the supply of labor inelastic. So the supply of skilled labor is inelastic and the supply of unskilled labor is elastic.

The individual supply curve (income-leisure trade-off)

The individual supply curve for labor can be explained with the help of income-leisure trade off. A rise in the wage rate implies a change in relative price of income and leisure. Income becomes cheaper relative to leisure, each hour worked earns more income than before or leisure becomes more expensive, since each hour of leisure consumed is at the expense of more income forgone.

This change in relative prices has both income and substitution effects. The substitution effect leads the individual to earn more income and less of more expensive leisure. But after achieving a certain level of income, a further increase in the wage rate may persuade the labor to enjoy greater income and a greater leisure time also. This is called the income effect of the wage increase and is achieved when the increase in wage rate leads to a fall in the number of hours worked, provided that the rise in wage rate is proportionately greater than the reduction in number of hours of work.



There is no such thing as luck, believe in hard work

Labor market

Assume that an individual can choose the number of hours to work and has 16 hours a day to divide between work and leisure. In the diagram, with an hourly wage rate of \$10, budget line B1 shows all the possible combinations of daily income and leisure hours. For e.g. at point X the individual has an income of \$100 by working 10 hours and taking 6 hours of leisure.

At an hourly rate of \$20 the budget line become B2 and the labor increases the hours of work or substituted the hours of leisure for income. He maximizes utility at point Y by working 14 hours and earns an income of \$280. Thus the higher wages have encouraged the individual to work more number of hours.

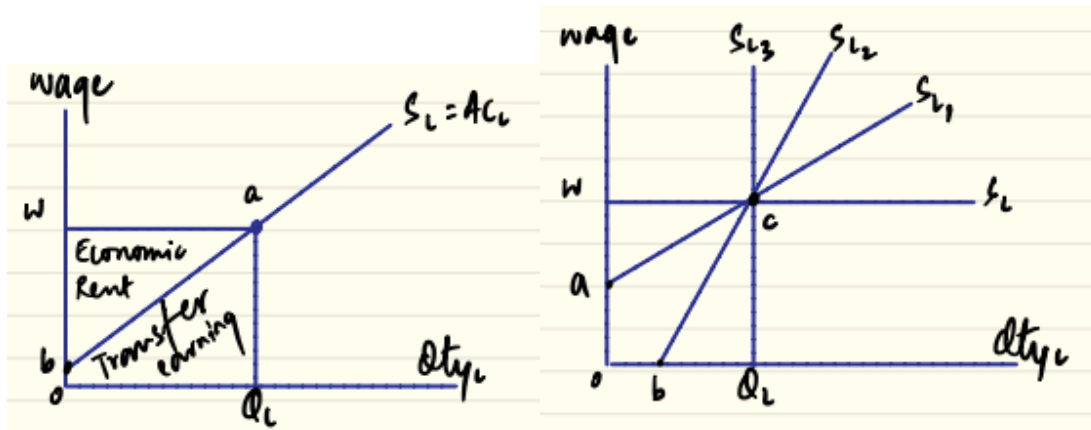
As the wage rate increases to \$30 per hour the budget line moves to B3 where the labor works for 12 hours and enjoys greater income of \$360 and greater leisure time or less hours of work. The income effect leads the individual to earn more income and more leisure provided that the reduced number of hours of work is proportionately less than the increased wage rate.

Because of the substitution effect the labor supply curve will be upward sloping and because of income effect the supply curve will be downward sloping. So up to some wage rate the number of hours of work and wage rate will positively related and after that the curve will bend backwards i.e. wage rate and hours worked will be negatively related.

ECONOMIC RENT and TRANSFER EARNINGS: The transfer earnings or the supply price of labor is the minimum payment required to prevent the labor transferring to another employer or another occupation. It is determined by what the labor could earn in its next best paid employment. Transfer earnings may be regarded as the opportunity cost of keeping labor in its present use, or it may be regarded as the labor's supply price in its present occupation. For example, suppose the minimum weekly wage which would persuade someone to work as a lorry driver is \$300, but he actually receives a wage of \$350 per week. We can say that his transfer earnings amount to \$300 per week and he is receiving \$50 per week in the form of economic rent. Economic rent therefore may be defined as any payment made to labor which is in excess of its transfer earnings.

There is no such thing as luck, believe in hard work

Labor market



The amount of economic rent depends on the elasticity of supply. The greater the elasticity of supply, the smaller the economic rent and the more inelastic the supply curve is the greater is the economic rent. Highly specialized labor is in very inelastic supply e.g. surgeons, physicist, highly qualified architects. Their transfer earnings are very small but economic rent is very high as shown in the figure below.

WAGE DETERMINATION IN MONOPSONY MARKET

A monopsonist is an employer with monopolistic buying (hiring) power. Monopsony has the following characteristics:

1. The firm's employment is a large portion of the total employment of a particular kind of labor
2. This type of labor is relatively immobile occupationally and geographically
3. The firm is a wage maker and that the wage rate it must pay varies directly with the number of workers it employs.

When a firm hires a considerable portion of the total available supply of a particular type of labor, its decision to employ more workers will affect the wage rate paid to that labor. The supply is in effect the average cost of labor curve; each point on it indicates the wage rate (cost) per worker which must be paid to attract the corresponding number of workers. But the higher wages needed to attract additional workers will also have to be paid to all workers

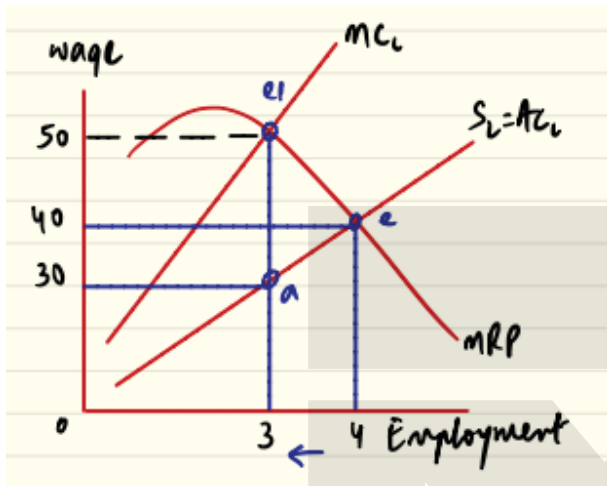
TABLE 1 A Monopsonist's Marginal Cost of Labor

Labor (number of workers)	Wage (per hour)	Total cost of labor	Marginal cost of labor
1	\$10	\$10	\$10
2	15	30	20
3	20	60	30
4	25	100	40
5	30	150	50

There is no such thing as luck, believe in hard work

Labor market

currently employed at lower wage rates. As for the cost, payment of a uniform wage to all workers mean that the cost of an extra worker - the marginal cost of labor (MCL) - will exceed the wage rate by the amount necessary to bring the wage rate of all workers up to the new wage level.



Equilibrium: to maximize profits the monopsonist will equate marginal cost of labor with the MRP. The number of labors hired by the monopsonist is indicated by 3 and the wage rate paid 30 is indicated by the corresponding point in the labor supply or average cost of labor curve, the firm needs not to pay a wage equal to MRP to attract 3 workers.

TRADE UNION are associations of workers formed for the purposes of improving the pay and working conditions of their members. Over the years, trade unions have played an important part in achieving better working conditions for the workers, as well as protection from unfair dismissal, the right to compensate for injury at work, the right to redundancy pay and many other benefits.

The individual worker is in a weak bargaining position and the main purpose of a trade union is to improve this weakness by forcing the employer to negotiate with representatives of the labor force.

Trade unions have two major functions, first, to bargain on behalf of their members for better pay and working conditions, and to persuade the government to pass legislation in favor of labor. Nowadays, unions have less power and influence in the labor market than they did two decades ago although in several big

There is no such thing as luck, believe in hard work

Labor market

industries they can still exert their industrial muscle. Power has gradually reduced for a variety of reasons:

1. **Employment legislation** which has outlawed illegal strikes, given employers the right to seek compensation for the effects of certain forms of industrial action and requires all unions to hold secret ballots of their members before any strike action is permitted.
2. The effects of **increase competition in product markets** - in nearly every domestic market for goods and services, there is greater competition than there was few years ago. Be it the intensity of global competition from lower cost producers or the deregulations of markets that has increased market contestability, trade unions have had to adjust to a world where the pricing power of manufacturers and service industries has been severely curtailed. Hence the increasing demands from businesses to link pay and conditions to worker productivity.
3. **Patterns of employment**: there has been a long term change in the structure of employment in many countries away from traditionally strong union sectors such as heavy engineering, coal mining, steel and textiles, towards service sector jobs in the private sector where union density is much lower. In situation of full employment when the union membership is very high and the union funds are in a healthy state, the unions are in a strong bargaining position but during unemployment they are weak.

With so many unions in an economy, it is almost impossible to classify them into clearly defined groups. It is possible, however, to distinguish four basic types of trade unions.

Craft union represents workers with particular skills, **industrial unions** represent all the workers in a particular industry, **general union** represents everyone wishing to join, and **white collar unions** represent administrative, clerical and other non-manual workers.

The bases of wage claims: trade unions demands for higher wages are normally based on one or more of four grounds:

The cost of living argument: When the prices increase, the real income of the labors fall and in order to maintain their real income, they demand increase in

There is no such thing as luck, believe in hard work

Labor market

wages. If the rise in the cost of living brought about by a previous wage increase then it seems difficult to justify the wage claim on purely economic grounds. The cost of living argument is one which attracts powerful social and political support and in recent years its acceptance has led to the widespread use of measures which link wages to the retail price index.

The differential argument: This argument rests on the widely held conviction that it is fair that workers doing similar jobs should get similar rewards. A wage settlement in one sector of the economy causes a chain reaction of wage claims in other sectors of the economy. If one sector is granted a wage increase it means that the existing pattern of wage differential has been upset. This will invariably provoke demands in other occupations for wage increases to restore the previous differentials.

The profitability argument: When the profits of firms increase, the labors also demand their share of profits in the form of increase in their wage rates. The individual union in pursuing the interests of its workers feels completely justified in pressing for an increase in wages whenever the profits in the industry are increasing.

The productivity argument: Improvements in labor productivity are widely accepted as justifiable reasons for increase in wages. Where increases in productivity result from the workers acceptance of new methods of production which impose greater strain, or responsibility, or call for retraining, claims for higher rewards are fully merited since the MRP of labor will increase.

How trade unions increase wages?

Trade unions raise wages of its members by either increasing the demand or restricting the supply of workers.

Increasing the demand for labor: From the union's viewpoint, the most desirable technique for rising wage rates to increase the demand for labor. An increase in the demand for labor will result in both higher wages and more jobs. The relative sizes of these increases will depend on the elasticity of labor supply. A union might increase labor demand by altering one or more determinants of labor demand i.e. by

There is no such thing as luck, believe in hard work

Labor market

increasing the demand for the product it is producing or by increasing the labor productivity.

Restricting the supply of labor: The union's bargaining power also depends upon their ability to control supply of labor to the industry. In cases where the union has established a closed shop, only union members may be employed and the union can effectively control the supply of labor. Trade union can also influence recruitment in other ways as by enforcing apprenticeship regulations and restricting employment to those who possess qualifications acceptable to the union.

Bilateral monopoly - this occurs when a monopoly buyer of labor faces a monopoly seller e.g. a monopsonist union. The outcome here depends on the bargaining strength of each side. Unions may be able to raise both the wage rate and employment

WAGE DIFFERENTIALS

The MRP theory suggests perfect competition in the labor market where all the labors are homogenous, so they are perfectly mobile among different occupations. It suggests that in an economy having one type labor and one type job, competition would result in a single wage rate for all workers. In reality wage rates differ in one or more of the following ways:

Incomes vary with the type of jobs: Dustmen earn less than a computer operator.

Incomes vary with education: Average earnings of people with university degrees exceed the average earnings of those with only A-levels.

Incomes vary with age: Average earnings rise until a persons' mid-40s and fall thereafter.

Incomes vary with years on the job: Generally, the longer one stays with one firm, the higher the income one earns.

Incomes vary with sex and race: Discrimination between blacks and whites, men and women, minority and majority groups, will lead to difference in earnings.

There is no such thing as luck, believe in hard work

Labor market

Incomes vary with the type of labor market in which it sells its services:

A labor working in the market dominated by the trade union earns more than the labor working in competitive market without union.

Differentials due to basic differences:

Non competing groups: Workers are not homogenous; they differ in their mental and physical capacities and in their education and training. At any point in time the labor force can be thought of as falling in to many non-competing groups. Few workers have the abilities to be brain surgeons, research chemists, entertainers, or professional athletes. The result is that the supplies of these types of labors are very small in relation to the demand for them and consequently wages and salaries are high. These groups do not compete with one another i.e. the wage rate is determined in all the labor markets independent of one another.

Differentials due to human capital: Human capital refers to the skills that a labor possesses. A human capital refers to expenditures on education and training which improve the skills and therefore, the productivity of workers.

Differentials due to labor market discrimination: Some forms of discrimination make it difficult or impossible for certain groups to take certain jobs, even if they are equipped by skills and education for these jobs. Discrimination reduces the supply of labor in some occupations, by keeping out the groups who are discriminated against, causing the wage rate in that occupation to increase. It also increases the supply in other occupations which are open to the groups subject to discrimination causing the wage rate in these occupations to fall.

Wage differential due to different labor market structures: In competitive labor market where firms and labors are wage takers, labors are homogenous, and occupationally mobile, equilibrium wage rate is determined at a point where demand and supply of labor are equal.

In a single wage monopsonist market, the monopsonist is a single buyer of a particular labor. In this market the employment is determined at the intersection of MC of labor and MRP and the wage rate is determined on the corresponding point on the supply curve. So, monopsony results in lower level of employment and

There is no such thing as luck, believe in hard work

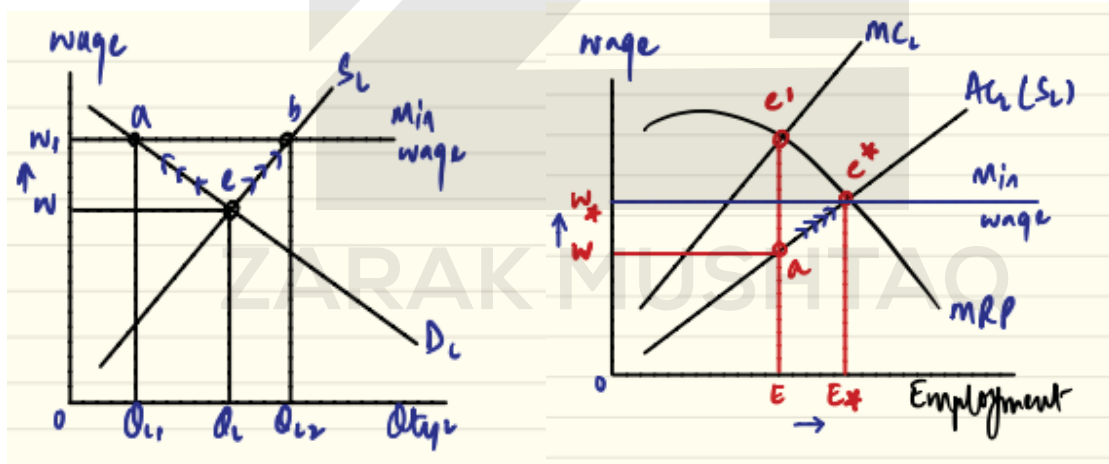
Labor market

lower wage rate than when labor is purchased competitively. A discriminating monopsonist makes a separate bargain from each group of labor or from each individual worker and pay just what is needed to persuade the worker to accept the job.

Differentials due to trade union action: Trade unions also play an important role in the determination of wage rates in different industries leading to difference in the wage rates. In industries where the unions are strong they will manage to achieve more frequent increases in the wage rates than the industries where they are either absent or are less influential. E.g. in the monopsony market, it can derive the wage rate above the competitive wage rate.

What if a union fixes a minimum wage?

A union minimum wage: If a trade union enters the labour market and becomes the monopoly supplier of labour, it can force the monopsonist to pay a wage at, or nearer to, the market rate, and employ more workers. At a minimum wage of £40, the supply of labour is horizontal at this wage, with the $MCL = ACL (S)$, and the profit maximizing monopsonist would employ up to the point where the $MCL = MRP$, which is at 4 workers - i.e. the market wage rate and the market level of employment.



However, if the minimum wage is set above £40, demand will contract and fewer will be employed. For example, setting the wage at £60 would mean only 2 hairdressers are employed.

The impact of the union minimum wage

There is no such thing as luck, believe in hard work

Labor market

The effect of the minimum wage clearly depends upon its level, and whether it is set above the market rate - the greater it is above the competitive market rate, the lower the level of employment. The impact of the union minimum wage also depends on the elasticity of demand and supply of labour. For example, if the demand for labour is relatively inelastic, jobs lost due to the minimum wage will be relatively small. Similarly, if supply is inelastic, the minimum rate may not result in a significant change in employment.



There is no such thing as luck, believe in hard work